

AGML Lab: Week 4

Naive Bayes

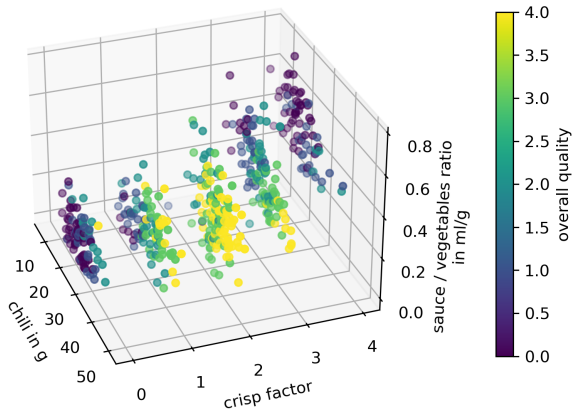


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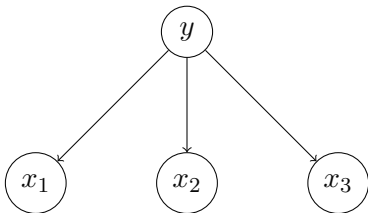
20.05.2026

(Jupyter Notebook)

- ▶ features might be discrete and continuous
- ▶ distributions of features might follow different models

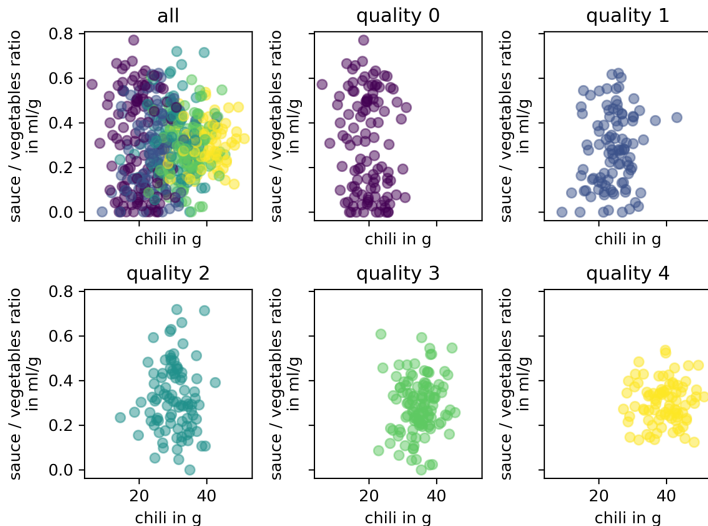


Naive Bayes Assumption! Features are conditionally independent given the class label.



$$p(x_1, x_2, x_3|y) = p(x_1|y)p(x_2|y)p(x_3|y)$$

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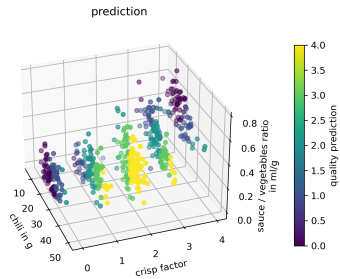
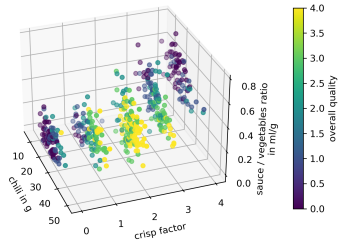


Now classifying is easy with Bayes theorem:

$$p(y|x) = \frac{\overbrace{p(x|y)}^{\text{likelihood}} \overbrace{p(y)}^{\text{prior}}}{\underbrace{p(x)}_{\text{evidence}}}$$

$$\begin{aligned}\arg \max_y p(y|x) &= \arg \max_y \frac{p(x|y)p(y)}{p(x)} \\ &= \arg \max_y p(x|y)p(y) \\ &= \arg \max_y \log p(x|y) + \log p(y) \\ &= \arg \max_y \log p(x_1|y) + \log p(x_2|y) \\ &\quad + \log p(x_3|y) + \log p(y)\end{aligned}$$

→ we just need to fit $p(x_i|y)$ for all features i and $p(y)$ (or the log-versions, whichever is more convenient)



(Jupyter Notebook)

1. Given the kebab dataset (discrete: 5 crispness levels, 5 quality levels; continuous (normal distribution): chili and sauce/vegetable ratio), how many values do we have to save for classification?
2. What changes for classification if we have a dynamic number of features? Example: text classification, every word can influence the probability of a sentence.